Md. Abu Bokkor Shiddik

Researcher | Bachelor of Science in Statistics

E-mail: abubokkor.brur60@gmail.com | Phone: +8801874001360

Location: Park Mor, Tajhat, Rangpur - 5400 , Bangladeh

Link: Portfolio | Linkdin | Google Scholar | ORCID | GitHub | Web of Science | Scopus ID

RESEARCH INTEREST

I am interested in applying data science, machine learning, and explainable AI to global health, with a focus on disease modeling, climate-health interactions, and interpretable decision support systems.

SKILLS

Programming and tools
(R, Python, C, Java, SPSS, LaTeX, Git, GitHub, HTML)

Data analysis and visualization
(ggplot2, dplyr, pandas, matplotlib, seaborn, Excel, PowerPoint)

Machine learning, deep learning and AI
(Linear regression, Logistic regression, Ridge regression, Lasso regression, Decision Trees, Random Forest, Gradient boosting: XGBoost, LightGBM, CatBoost, Support Vector Machines, K-Nearest Neighbors, Naive Bayes, Clustering: K-means, Hierarchical, DBSCAN, Dimensionality reduction: PCA, Artificial Neural Networks, Deep Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, LSTM, GRU, Spatio-temporal deep learning: ConvLSTM, STCNN, STGNN, Bayesian modeling, Probabilistic approaches)

Explainable AI and model interpretation
(SHAP, LIME, Integrated Gradients, Layer-wise Relevance Propagation, Feature importance analysis)

Statistical and econometric methods
(Hypothesis testing, ANOVA, Multivariate analysis, Time series analysis: ARIMA, SARIMA, Bayesian inference, Generalized linear models)

Geospatial and spatio-temporal modeling
(GIS: sf, raster, Spatial analysis, Spatio-temporal modeling, Hotspot analysis: Getis-Ord Gi*)

Public and global health analytics
(Epidemiological modeling, Disease prediction, Climate-health data integration, Health risk assessment, Early warning systems)

Research and academic skills
(Scientific writing, Systematic review, Literature synthesis, Manuscript preparation, Peer review, Data interpretation, Reproducible research)

EDUCATION

March, 2022 -February, 2026	**Bachelor of Science in Statistics** Begum Rokeya University, Rangpur, Bangladesh
January, 2018 - December, 2019	**Higher Secondary Certificate (Science)** Police Lines School and College, Rangpur, Bangladesh

COURSES AND TRAINING

21 March, 2026	**Gen AI use in the research workflow** A module by Researcher Academy, ELSEVIER
14 January, 2026	**How to write for an interdisciplinary audience** A module by Researcher Academy, ELSEVIER
14 January, 2026	**How to write case reports** A module by Researcher Academy, ELSEVIER
12 January, 2026	**How to identify research gaps** A module by Researcher Academy, ELSEVIER
11 January, 2026	**Beginners' guide to writing a manuscript in LaTeX** A module by Researcher Academy, ELSEVIER
10 January, 2026	**How to write and review a methods article** A module by Researcher Academy, ELSEVIER
10 January, 2026	**How to write and review a methods article** A module by Researcher Academy, ELSEVIER
10 January, 2026	**An editor's guide to writing a review article** A module by Researcher Academy, ELSEVIER
15 March, 2025	**Practical Time Series Analysis** An online course by The State University of New York
10 September, 2024	**Introduction to Climate Risk Informed Decision Analysis (CRIDA)** a course of study offered by United Nations Educational, Scientific and Cultural Organization (UNESCO)
7 June, 2024	**Python 101 for Data Science** PY0101EN, provided by IBM
10 March, 2024	**Successful Presentation** An online course by University of Colorado Boulder
16 January, 2024	**Data Visualization with R** DV0151EN, provided by IBM
16 January, 2024	**Predictions: Regression for Car Mileage and Diamond Price** GPXX0AEIEN, provided by IBM Developer Skills Network
4 January, 2024	**Machine Learning with Python** ML0101EN, provided by IBM
December, 2023	**Interdisciplinary Research Methodology Course for Beginners** by Research Society-Begum Rokeya University (RSBRU)
30 December, 2023	**R for Data Science** RP0101EN, provided by IBM
29 December, 2023	**Data Science Methodology** DS0103EN, provided by IBM
26 December, 2023	**Research Methodologies** An online course by Queen Mary University of London

23 December, 2023	**Introduction to Computer Programming** An online course by University of London
24 December, 2023	**Introduction to Statistical Analysis: Hypothesis Testing** By SAS
19 December, 2023	**Introduction to Data Analytics** An online course by IBM
4 December, 2023	**Machine Learning for All** An online course by University of London
6 November, 2023	**Education in Emergencies Global Course** provided by UNICEF
24 November, 2023	**What is Data Science?** An online course by IBM
18 November, 2023	**Understanding Research Methods** An online course by University of London
15 November, 2023	**Basic Python Course** By Cambridge International Qualifications (CIQ)
1 November, 2023	**Leadership for Planning and Decision-Making with Adolescents** By UNICEF
July, 2016-December, 2016	**Deploma in Computer Science** A six month training course by Bangladesh Computer Education (BCE)

RESEARCH PROJECT

Project Title: Exploring community knowledge, health belief and healthcare-seeking behaviours regarding Non-Communicable Diseases (NCD's) in Bangladesh

Funding Source: University Grant Commission

Role: Research Assistant (Start date: Oct.2025 & end date: Feb.2026) , Department of Statistics, Begum Rokeya University, Rangpur

Project Title: District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning

Description: A project submitted in partial fulfillment of the requirements for the degree of B.Sc. In Statistics, Begum Rokeya University, Rangpur.

RESEARCH & PUBLICATION

1. Rahman, M.S., **Shiddik, A.B.** Utilizing artificial intelligence to predict and analyze socioeconomic, environmental, and healthcare factors driving tuberculosis globally. *Sci Rep* 15, 13619 (2025). https://doi.org/10.1038/s41598-025-96973-w

2. Rahman, M. S., Amrin, M., & **Bokkor Shiddik, M. A.** (2025). Dengue Early Warning System and Outbreak Prediction Tool in Bangladesh Using Interpretable Tree-Based Machine Learning Model. *Health Science Reports*, *8*(5), e70726. https://doi.org/10.1002/hsr2.70726

3. Rahman, M. S., & **Shiddik, M. A. B**. (2025). Explainable artificial intelligence for predicting dengue outbreaks in Bangladesh using eco-climatic triggers. *Global Epidemiology*, 100210.

4. Rahman, M. S., & **Shiddik, M. A. B.** (2025). Unraveling global malaria incidence and mortality using machine learning and artificial intelligence–driven spatial analysis. *Scientific Reports*, *15*(1), 28334.

5. Rahman, S., & **Shiddik, A. B**. (2025). Reflections on explainable artificial intelligence for predicting dengue outbreaks in Bangladesh. *Global Epidemiology*, 100230.

6. Rahman, M. S., & Shiddik, M. A. B. (2025). Data-Driven Dengue Prevention Strategies in Bangladesh using Explainable Artificial Intelligence and Causal Inference. *International Journal of Statistical Sciences*, *25*(2), 101-113.

7. Rahman, M. S., & **Shiddik, M. A. B.** (2025). Leveraging explainable artificial intelligence and spatial analysis for communicable diseases in Asia (2000–2022) based on health, climate, and socioeconomic factors. *International Journal of Health Geographics*, *24*(1), 45.

8. Rahman, S. and **Shiddik, A.B.**, 2026. Bayesian spatiotemporal modelling of neonatal, infant and under-5 mortality (2000–2022) in 41 Asian countries: a population-level observational study. *BMJ open*, *16*(2), p.e108632.

9. **Shiddik, M.A.B.**, Toshi, F.Z., Yesmin, S. and Rahman, M.S., 2026. District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning. *Tropical Medicine and Infectious Disease*, 11(3), p.73.

REVIEWER SERVICE

- **BMJ global health** (1 Article in 2026)
- **BMJ public health** (1 Article in 2026)
- **Health science reports** (1 Article in 2026, 2 Article in 2025)
- **Journal of mathematics and statistics** (1 Article in 2025)
- **PLOS digital health** (6 Article in 2026, 6 Article in 2025)

DIGITAL PROJECTS AND TOOLS

whatdis() — R distribution detector

Developed whatdis(), a custom R package for checking and visualizing data distribution. Code and documentation available on GitHub: https://github.com/abubokkorshiddik/R-distribution-detector

Dengue analytics dashboard, Bangladesh

Built an interactive Dengue dashboard that shows patterns and insights for district level dengue cases and early warning modeling in Bangladesh. Live demo: https://abubokkorshiddik.github.io/dengue-dashboard

Snake game

Created a classic Snake game using JavaScript and CSS with interactive controls and scoring. Live demo: https://abubokkorshiddik.github.io/snake-game

Systematic review dashboard

Built an interactive Systematic review dashboard for searching, screening, visualizing trends, filtering by author/title/DOI/institution, showing abstracts, highlighting top authors, and exporting results for systematic reviews. Live demo: https://abubokkorshiddik.github.io/research-explorer

CONFERENCE & WORKSHOP

International Conference on Applied Statistics and Data Science-2025
University of Dhaka

Abstract ID: P-145 & P-105
Presented research work: Spatio-temporal modeling of global tuberculosis prediction based on unified-machine learning framwork

Workshop on 'Promotion of Research in Higher Education (10 march, 2025)
Institutional Quality Assurance Cell (IQAC), Begum Rokeya University, Rangpur.

CERTIFICATIONS

March, 2026	**Certificate of Publication** The certificate of publication for the article titled:District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning; *Trop. Med. Infect. Dis.* 2026, Volume 11, Issue 3, 73
March, 2026	**Certificate of Excellence; Research design** by Researcher Academy, ELSEVIER
23 March 2026	**Certificate of Appreciation in recognition of peer reviewer contributions for BMJ Global Health** by BMJ
16 February 2026	**Certificate of Appreciation in recognition of peer reviewer contributions for BMJ Public Health** by BMJ
January, 2026	**Certificate of Excellence; Technical writing skills** by Researcher Academy, ELSEVIER
December, 2025	**Certificate of Appreciation for presenting a research paper at the Poster session of the International Conference on Applied Statistics and Data Science (ICASDS)** by Institute of Statistical Research and Training (ISRT), University of Dhaka
June, 2022	**Certificate of Appreciation for participating in "Art Exhibition"** by Film and Art Society (FLAS), Begum Rokeya University, Rangpur

ACADEMIC CONTENT AND OUTREACH

Created educational video tutorials on Basic Statistics, R and Python for data analysis
YouTube channel: https://www.youtube.com/@abubokkorshiddik.abs798

Shared public health and data science resources to promote research skills among students and early-career researchers

REFERENCE

Dr. Md. Siddikur Rahman
Department of Statistics,
Begum Rokeya University,Rangpur, Bangladesh.
E-mail: siddikur@brur.ac.bd